# SYNAPSE API Specification

Comprehensive API Documentation for AI-Powered Technology Intelligence Platform

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# Chapter 1

# Introduction

## 1.1 Overview

SYNAPSE is a FAANG-style AI-powered technology intelligence platform built with FastAPI and Django REST framework. This document provides comprehensive documentation for all API endpoints, authentication mechanisms, and integration guidelines.

The SYNAPSE platform enables users to:

- Access curated technology articles and research papers
- Search across multiple content sources using semantic search
- Interact with AI assistants for insights and summarization
- Manage automated workflows and agent tasks
- Track technology trends and create custom collections
- Generate comprehensive reports and documents

## 1.2 API Base URL

All API endpoints are prefixed with the base URL:

```
https://api.synapse.ai/api/v1
```

## 1.3 API Versioning

The SYNAPSE API uses URL-based versioning. The current version is `v1`. Future versions will be introduced with breaking changes while maintaining backward compatibility for at least one major version.

Version information is included in the URL path: `/api/v1/...`

# Chapter 2

# Authentication

## 2.1 JWT Bearer Tokens

SYNAPSE uses JWT (JSON Web Tokens) for authentication. All authenticated endpoints require an `Authorization` header with a Bearer token.

## 2.2 Token Structure

The system maintains two types of tokens:

- **Access Token**: Short-lived token (15 minutes expiration) used for API requests
- **Refresh Token**: Long-lived token (7 days expiration) used to obtain new access tokens

## 2.3 Token Refresh Strategy

The recommended token refresh strategy is:

1. Client stores both access and refresh tokens upon successful login
2. Client uses access token for API requests
3. When access token expires (401 Unauthorized), client uses refresh token to obtain a new access token
4. If refresh token is also expired, user must re-authenticate

## 2.4 Authorization Header Format

```
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

# Chapter 3

# Rate Limiting

## 3.1 Rate Limit Tiers

SYNAPSE implements rate limiting based on subscription tier:

| Tier | Requests per Minute | Monthly Limit |
|---|---|---|
| Free | 100 | 100,000 |
| Premium | 1,000 | 5,000,000 |
| Enterprise | Unlimited | Unlimited |

## 3.2 Rate Limit Headers

Each response includes rate limit information:

- `X-RateLimit-Limit`: Maximum requests allowed per minute
- `X-RateLimit-Remaining`: Requests remaining in current window
- `X-RateLimit-Reset`: Unix timestamp when the rate limit resets

# Chapter 4

# Error Response Format

## 4.1 Standard Error Envelope

All errors are returned in a consistent format:

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "One or more fields are invalid",
    "details": [
      {
        "field": "email",
        "message": "Invalid email format"
      }
    ]
  }
}
```

## 4.2 HTTP Status Codes

| Status Code | Description |
|---|---|
| 200 | OK - Request succeeded |
| 201 | Created - Resource successfully created |
| 204 | No Content - Request succeeded, no response body |
| 400 | Bad Request - Invalid request parameters |
| 401 | Unauthorized - Missing or invalid authentication |
| 403 | Forbidden - Authenticated but not authorized for resource |
| 404 | Not Found - Resource does not exist |
| 422 | Unprocessable Entity - Request validation failed |
| 429 | Too Many Requests - Rate limit exceeded |
| 500 | Internal Server Error - Server error |
| 503 | Service Unavailable - Server temporarily unavailable |

# Chapter 5

# Pagination

## 5.1 Pagination Format

List endpoints support pagination with the following query parameters:

- `page`: Page number (1-indexed, default: 1)
- `page_size`: Number of results per page (default: 20, max: 100)

## 5.2 Paginated Response

```
{
  "success": true,
  "data": {
    "page": 1,
    "page_size": 20,
    "total": 1250,
    "total_pages": 63,
    "results": [
      { /* items */ }
    ]
  }
}
```

# Chapter 6

# Authentication Endpoints

## 6.1 POST /auth/register

**Description**: Register a new user account

**Headers**:

```
Content-Type: application/json
```

**Request Body**:

```
{
  "email": "user@example.com",
  "password": "SecurePassword123!",
  "first_name": "John",
  "last_name": "Doe",
  "company": "Tech Corp",
  "job_title": "AI Engineer"
}
```

**Response (201 Created)**:

```
{
  "success": true,
  "data": {
    "id": "usr_12345abcde",
    "email": "user@example.com",
    "first_name": "John",
    "last_name": "Doe",
    "created_at": "2024-01-15T10:30:00Z"
  }
}
```

**Status Codes**:

- 201: User successfully created
- 400: Invalid request data
- 422: Email already exists or validation failed

## 6.2 POST /auth/login

**Description**: Authenticate user and return JWT tokens

**Request Body**:

```
{
  "email": "user@example.com",
  "password": "SecurePassword123!"
}
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "token_type": "bearer",
    "expires_in": 900,
    "user": {
      "id": "usr_12345abcde",
      "email": "user@example.com",
      "first_name": "John",
      "last_name": "Doe"
    }
  }
}
```

**Status Codes**:

- 200: Login successful
- 400: Invalid credentials
- 404: User not found

## 6.3 POST /auth/logout

**Description**: Logout and invalidate current token

**Headers**:

```
Authorization: Bearer <access_token>
```

**Request Body**:

```
{
  "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

**Response (204 No Content)**:

```
(empty response body)
```

**Status Codes**:

- 204: Successfully logged out
- 401: Unauthorized

## 6.4 POST /auth/token/refresh

**Description**: Refresh expired access token using refresh token

**Request Body**:

```
{
  "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "token_type": "bearer",
    "expires_in": 900
  }
}
```

**Status Codes**:

- 200: Token refreshed successfully
- 401: Invalid or expired refresh token

## 6.5 POST /auth/password/reset

**Description**: Request password reset link via email

**Request Body**:

```
{
  "email": "user@example.com"
}
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "message": "Password reset email sent",
    "email": "user@example.com"
  }
}
```

**Status Codes**:

- 200: Reset email sent
- 404: User not found

## 6.6 POST /auth/password/reset/confirm

**Description**: Confirm password reset with token and new password

**Request Body**:

```
{
  "token": "reset_token_from_email",
  "password": "NewPassword456!"
}
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "message": "Password successfully reset"
  }
}
```

**Status Codes**:

- 200: Password reset successful
- 400: Invalid or expired token
- 422: Password validation failed

## 6.7 GET /auth/me

**Description**: Get current authenticated user profile

**Headers**:

```
Authorization: Bearer <access_token>
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "usr_12345abcde",
    "email": "user@example.com",
    "first_name": "John",
    "last_name": "Doe",
    "company": "Tech Corp",
    "subscription_tier": "premium",
    "created_at": "2024-01-15T10:30:00Z"
  }
}
```

**Status Codes**:

- 200: Profile retrieved successfully
- 401: Unauthorized

## 6.8 PUT /auth/me

**Description**: Update current user profile

**Headers**:

```
Authorization: Bearer <access_token>
Content-Type: application/json
```

**Request Body**:

```
{
  "first_name": "John",
  "last_name": "Smith",
  "company": "New Tech Corp",
  "job_title": "Senior AI Engineer"
}
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "usr_12345abcde",
    "email": "user@example.com",
    "first_name": "John",
    "last_name": "Smith",
    "company": "New Tech Corp",
    "job_title": "Senior AI Engineer"
  }
}
```

**Status Codes**:

- 200: Profile updated successfully
- 401: Unauthorized
- 422: Validation failed

# Chapter 7

# Articles Endpoints

## 7.1 GET /articles/

**Description**: List articles with pagination and filtering

**Headers**:

```
Authorization: Bearer <access_token>
```

**Query Parameters**:

- `page`: Page number (default: 1)
- `page_size`: Results per page (default: 20, max: 100)
- `topic`: Filter by topic (e.g., machine_learning)
- `source`: Filter by source (e.g., arxiv, medium)
- `date_from`: Start date (ISO 8601)
- `date_to`: End date (ISO 8601)
- `trending`: Boolean to get trending articles

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "page": 1,
    "page_size": 20,
    "total": 5423,
    "results": [
      {
        "id": "art_abc123",
        "title": "Advances in Transformer Architecture",
        "summary": "Recent developments in transformer
           models...",
        "source": "arxiv",
        "topic": "machine_learning",
        "published_at": "2024-01-20T10:00:00Z",
```

```
        "url": "https://arxiv.org/...",
        "read_time_minutes": 12
      }
    ]
  }
}
```

**Status Codes**: 200, 401, 422

## 7.2 GET /articles/{id}

**Description**: Get detailed article content

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "art_abc123",
    "title": "Advances in Transformer Architecture",
    "summary": "Recent developments in transformer models...",
    "content": "Full HTML content of the article...",
    "source": "arxiv",
    "topic": "machine_learning",
    "author": "John Researcher",
    "published_at": "2024-01-20T10:00:00Z",
    "url": "https://arxiv.org/...",
    "views": 1250,
    "bookmarked": true,
    "related_articles": [
      {"id": "art_def456", "title": "..."}
    ]
  }
}
```

## 7.3 GET /articles/trending

**Description**: Get trending articles

**Query Parameters**:

- `period`: Time period (24h, 7d, 30d, default: 7d)
- `limit`: Number of results (default: 10, max: 50)

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "period": "7d",
    "results": [
      {
```

```
        "id": "art_abc123",
        "title": "Trending Article Title",
        "trend_score": 95.5,
        "view_count": 5000
      }
    ]
  }
}
```

## 7.4 GET /articles/topics

**Description**: Get all available article topics

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "topics": [
      {
        "name": "machine_learning",
        "label": "Machine Learning",
        "article_count": 2341
      },
      {
        "name": "cloud_computing",
        "label": "Cloud Computing",
        "article_count": 1876
      }
    ]
  }
}
```

## 7.5 POST /articles/{id}/bookmark

**Description**: Bookmark an article

**Headers**:

```
Authorization: Bearer <access_token>
```

**Response (201 Created)**:

```
{
  "success": true,
  "data": {
    "article_id": "art_abc123",
    "bookmarked": true,
    "created_at": "2024-01-20T10:30:00Z"
  }
}
```

## 7.6 DELETE /articles/{id}/bookmark

**Description**: Remove article bookmark

**Response (204 No Content)**

## 7.7 GET /articles/search

**Description**: Keyword search across articles

**Query Parameters**:

- `q`: Search query (required)
- `page`: Page number (default: 1)
- `page_size`: Results per page (default: 20)

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "query": "transformers",
    "page": 1,
    "total": 342,
    "results": [
      {
        "id": "art_abc123",
        "title": "...",
        "excerpt": "...transformer models...",
        "relevance_score": 0.95
      }
    ]
  }
}
```

# Chapter 8

# Semantic Search

## 8.1 POST /search/semantic

**Description**: Perform semantic/vector search across all content

**Headers**:

```
Authorization: Bearer <access_token>
Content-Type: application/json
```

**Request Body**:

```
{
  "query": "How are transformers improving NLP?",
  "limit": 20,
  "filters": {
    "content_type": ["articles", "papers"],
    "source": "arxiv",
    "date_from": "2024-01-01",
    "date_to": "2024-01-31"
  }
}
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "query": "How are transformers improving NLP?",
    "results": [
      {
        "id": "art_abc123",
        "type": "article",
        "title": "Advances in Transformer Architecture",
        "summary": "...",
        "similarity_score": 0.92,
        "source": "arxiv",
        "published_at": "2024-01-20T10:00:00Z"
      }
    ],
    "execution_time_ms": 145
```

```
  }
}
```

**Status Codes**: `200`, `401`, `422`

# Chapter 9

# Research Papers Endpoints

## 9.1 GET /papers/

**Description**: List research papers with filtering

**Query Parameters**:

- `page`: Page number (default: 1)
- `page_size`: Results per page (default: 20)
- `category`: Filter by category (theoretical, applied, experimental)
- `difficulty`: Filter by difficulty (beginner, intermediate, advanced)
- `date_from`: Start date (ISO 8601)

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "page": 1,
    "total": 8932,
    "results": [
      {
        "id": "pap_xyz789",
        "title": "Attention Is All You Need",
        "authors": ["Vaswani, A.", "Shazeer, N."],
        "abstract": "The dominant sequence transduction...",
        "category": "theoretical",
        "difficulty": "advanced",
        "published_date": "2017-06-12",
        "citation_count": 45000,
        "pdf_url": "https://arxiv.org/pdf/..."
      }
    ]
  }
}
```

## 9.2 GET /papers/{id}

**Description**: Get paper detail with full metadata

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "pap_xyz789",
    "title": "Attention Is All You Need",
    "authors": ["Vaswani, A.", "Shazeer, N."],
    "abstract": "...",
    "category": "theoretical",
    "difficulty": "advanced",
    "published_date": "2017-06-12",
    "citation_count": 45000,
    "pdf_url": "https://arxiv.org/pdf/...",
    "key_concepts": ["attention", "transformers", "NLP"],
    "bookmarked": false
  }
}
```

## 9.3 GET /papers/trending

**Description**: Get trending research papers

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "period": "7d",
    "results": [
      {
        "id": "pap_xyz789",
        "title": "...",
        "citations_this_period": 234
      }
    ]
  }
}
```

## 9.4 POST /papers/{id}/bookmark

**Description**: Bookmark a research paper

**Response (201 Created)**

# Chapter 10

# Repositories Endpoints

## 10.1 GET /repos/

**Description**: List code repositories with filtering

**Query Parameters**:

- `language`: Filter by programming language
- `topic`: Filter by topic tag
- `stars_min`: Minimum GitHub stars
- `sort`: Sort by (stars, forks, recent, default: stars)

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "page": 1,
    "total": 12450,
    "results": [
      {
        "id": "repo_123abc",
        "name": "transformers",
        "owner": "huggingface",
        "url": "https://github.com/huggingface/transformers",
        "description": "State-of-the-art NLP library",
        "language": "Python",
        "stars": 95000,
        "forks": 18000,
        "topics": ["nlp", "transformers", "hugging-face"],
        "last_updated": "2024-01-20T15:30:00Z"
      }
    ]
  }
}
```

## 10.2 GET /repos/{id}

**Description**: Get repository details

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "repo_123abc",
    "name": "transformers",
    "owner": "huggingface",
    "url": "https://github.com/huggingface/transformers",
    "description": "State-of-the-art NLP library",
    "language": "Python",
    "stars": 95000,
    "forks": 18000,
    "open_issues": 234,
    "topics": ["nlp", "transformers"],
    "readme": "Markdown content of README...",
    "license": "Apache-2.0",
    "bookmarked": true
  }
}
```

## 10.3 GET /repos/trending

**Description**: Get trending repositories

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "period": "7d",
    "results": [
      {
        "id": "repo_123abc",
        "name": "...",
        "stars_gained_this_period": 500
      }
    ]
  }
}
```

## 10.4 POST /repos/{id}/bookmark

**Description**: Bookmark a repository

**Response (201 Created)**

# Chapter 11

# Videos Endpoints

## 11.1 GET /videos/

**Description**: List technology videos and tutorials

**Query Parameters**:

- `page`: Page number
- `topic`: Filter by topic
- `duration`: Filter by video length (short, medium, long)

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "page": 1,
    "total": 3421,
    "results": [
      {
        "id": "vid_456def",
        "title": "Transformers Explained",
        "channel": "3Blue1Brown",
        "duration_seconds": 1245,
        "published_at": "2024-01-15T10:00:00Z",
        "view_count": 125000,
        "topic": "machine_learning",
        "youtube_id": "ABC123XYZ"
      }
    ]
  }
}
```

## 11.2 GET /videos/{id}

**Description**: Get video details with transcript

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "vid_456def",
    "title": "Transformers Explained",
    "channel": "3Blue1Brown",
    "duration_seconds": 1245,
    "description": "In this video...",
    "transcript": "Timestamp: 00:00\nTranscript text...",
    "topic": "machine_learning",
    "youtube_id": "ABC123XYZ",
    "bookmarked": false
  }
}
```

## 11.3 GET /videos/trending

**Description**: Get trending videos

**Response (200 OK)**

# Chapter 12

# AI Assistant Endpoints

## 12.1 POST /ai/chat

**Description**: Send message to AI assistant and receive intelligent response

**Headers**:

```
Authorization: Bearer <access_token>
Content-Type: application/json
```

**Request Body**:

```
{
  "message": "Explain the concept of attention mechanisms",
  "conversation_id": "conv_abc123",
  "context": {
    "focus_areas": ["transformers", "NLP"],
    "difficulty_level": "intermediate"
  }
}
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "response": "Attention mechanisms allow models to...",
    "sources": [
      {
        "title": "Attention Is All You Need",
        "url": "https://arxiv.org/...",
        "type": "paper",
        "relevance": 0.95
      }
    ],
    "conversation_id": "conv_abc123",
    "response_time_ms": 2340
  }
}
```

**Status Codes**: 200, 401, 422

## 12.2 GET /ai/chat/{conversation_id}/history

**Description**: Get chat history for a conversation

**Headers**:

```
Authorization: Bearer <access_token>
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "conversation_id": "conv_abc123",
    "created_at": "2024-01-20T10:00:00Z",
    "messages": [
      {
        "id": "msg_123",
        "role": "user",
        "content": "Explain attention mechanisms",
        "timestamp": "2024-01-20T10:00:00Z"
      },
      {
        "id": "msg_124",
        "role": "assistant",
        "content": "Attention mechanisms allow...",
        "timestamp": "2024-01-20T10:00:30Z"
      }
    ]
  }
}
```

## 12.3 DELETE /ai/chat/{conversation_id}

**Description**: Clear/delete a conversation

**Headers**:

```
Authorization: Bearer <access_token>
```

**Response (204 No Content)**

## 12.4 POST /ai/summarize

**Description**: Request AI to summarize content

**Request Body**:

```
{
  "content_type": "article",
  "content_id": "art_abc123"
}
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "content_id": "art_abc123",
    "summary": "This article discusses...",
    "key_points": [
      "First key point",
      "Second key point"
    ],
    "character_count": 1245
  }
}
```

## 12.5 POST /ai/explain

**Description**: Get AI explanation of research paper or concept

**Request Body**:

```
{
  "type": "paper",
  "id": "pap_xyz789",
  "target_audience": "intermediate"
}
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "pap_xyz789",
    "explanation": "This paper introduces the concept of...",
    "key_concepts": ["attention", "transformers"],
    "learning_resources": [
      {
        "title": "Understanding Attention",
        "url": "https://..."
      }
    ]
  }
}
```

# Chapter 13

# Agent Tasks Endpoints

## 13.1 POST /agents/tasks

**Description**: Create an agent task for automated research

**Headers**:

```
Authorization: Bearer <access_token>
Content-Type: application/json
```

**Request Body**:

```
{
  "task_type": "research",
  "prompt": "Find latest papers on quantum computing",
  "parameters": {
    "date_range": "last_30_days",
    "min_citations": 10,
    "include_preprints": true
  }
}
```

**Response (201 Created)**:

```
{
  "success": true,
  "data": {
    "id": "task_12345",
    "task_type": "research",
    "status": "pending",
    "created_at": "2024-01-20T10:00:00Z",
    "estimated_completion": "2024-01-20T10:15:00Z"
  }
}
```

## 13.2 GET /agents/tasks

**Description**: List user's agent tasks

**Query Parameters**:

- `status`: Filter by status (pending, running, completed, failed)
- `page`: Page number

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "page": 1,
    "total": 24,
    "results": [
      {
        "id": "task_12345",
        "task_type": "research",
        "status": "completed",
        "created_at": "2024-01-20T10:00:00Z",
        "completed_at": "2024-01-20T10:15:00Z"
      }
    ]
  }
}
```

## 13.3 GET /agents/tasks/{id}

**Description**: Get task status and result

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "task_12345",
    "task_type": "research",
    "status": "completed",
    "prompt": "Find latest papers on quantum computing",
    "progress": 100,
    "created_at": "2024-01-20T10:00:00Z",
    "completed_at": "2024-01-20T10:15:00Z",
    "result": {
      "papers_found": 45,
      "items": [
        {
          "id": "pap_xyz789",
          "title": "...",
          "relevance_score": 0.98
        }
      ]
    }
  }
}
```

## 13.4 DELETE /agents/tasks/{id}

**Description**: Cancel a running task
**Response (204 No Content)**

## 13.5 GET /agents/tools

**Description**: List available agent tools
**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "tools": [
      {
        "name": "research_papers",
        "description": "Search and filter research papers",
        "parameters": {
          "query": "string",
          "date_range": "string"
        }
      },
      {
        "name": "github_search",
        "description": "Search GitHub repositories",
        "parameters": {
          "query": "string",
          "language": "string"
        }
      }
    ]
  }
}
```

# Chapter 14

# Document Generation Endpoints

## 14.1 POST /documents/generate

**Description**: Generate comprehensive documents (PDF, PPT, Word, Markdown)

**Headers**:

```
Authorization: Bearer <access_token>
Content-Type: application/json
```

**Request Body**:

```
{
  "doc_type": "pdf",
  "topic": "Machine Learning in 2024",
  "prompt": "Create a comprehensive report on ML trends",
  "include_sections": [
    "introduction",
    "market_analysis",
    "key_technologies",
    "case_studies"
  ]
}
```

**Response (201 Created)**:

```
{
  "success": true,
  "data": {
    "id": "doc_789xyz",
    "doc_type": "pdf",
    "title": "Machine Learning in 2024",
    "status": "generating",
    "created_at": "2024-01-20T10:00:00Z",
    "estimated_completion": "2024-01-20T10:30:00Z"
  }
}
```

## 14.2 GET /documents/

**Description**: List user's generated documents

**Query Parameters**:

- `page`: Page number
- `status`: Filter by status (generating, completed, failed)

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "page": 1,
    "total": 15,
    "results": [
      {
        "id": "doc_789xyz",
        "title": "Machine Learning in 2024",
        "doc_type": "pdf",
        "status": "completed",
        "created_at": "2024-01-20T10:00:00Z",
        "file_size_bytes": 2145000
      }
    ]
  }
}
```

## 14.3 GET /documents/{id}

**Description**: Get document metadata

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "doc_789xyz",
    "title": "Machine Learning in 2024",
    "doc_type": "pdf",
    "status": "completed",
    "created_at": "2024-01-20T10:00:00Z",
    "page_count": 42,
    "file_size_bytes": 2145000,
    "download_url": "https://..."
  }
}
```

## 14.4 GET /documents/{id}/download

**Description**: Download generated document file
**Response**: File binary stream

## 14.5 DELETE /documents/{id}

**Description**: Delete a generated document
**Response (204 No Content)**

# Chapter 15

# Automation Workflows Endpoints

## 15.1 POST /workflows/

**Description**: Create automation workflow

**Headers**:

```
Authorization: Bearer <access_token>
Content-Type: application/json
```

**Request Body**:

```
{
  "name": "Weekly ML Research Digest",
  "description": "Gather trending ML papers weekly",
  "trigger": {
    "type": "schedule",
    "cron": "0 9 * * 1"
  },
  "actions": [
    {
      "type": "search",
      "params": {
        "query": "machine learning",
        "date_range": "7d"
      }
    },
    {
      "type": "summarize",
      "params": {
        "max_items": 10
      }
    },
    {
      "type": "send_email",
      "params": {
        "template": "digest"
      }
    }
  ]
}
```

**Response (201 Created)**:

```
{
  "success": true,
  "data": {
    "id": "wf_123abc",
    "name": "Weekly ML Research Digest",
    "enabled": true,
    "created_at": "2024-01-20T10:00:00Z"
  }
}
```

## 15.2 GET /workflows/

**Description**: List user's workflows

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "page": 1,
    "total": 8,
    "results": [
      {
        "id": "wf_123abc",
        "name": "Weekly ML Research Digest",
        "enabled": true,
        "created_at": "2024-01-20T10:00:00Z",
        "last_run": "2024-01-19T09:00:00Z"
      }
    ]
  }
}
```

## 15.3 GET /workflows/{id}

**Description**: Get workflow detail

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "wf_123abc",
    "name": "Weekly ML Research Digest",
    "description": "...",
    "enabled": true,
    "trigger": {"type": "schedule", "cron": "0 9 * * 1"},
    "actions": [{"type": "search", "params": {}}],
```

```
    "created_at": "2024-01-20T10:00:00Z",
    "updated_at": "2024-01-20T10:00:00Z"
  }
}
```

## 15.4 PUT /workflows/{id}

**Description**: Update workflow
**Response (200 OK)**

## 15.5 DELETE /workflows/{id}

**Description**: Delete workflow
**Response (204 No Content)**

## 15.6 POST /workflows/{id}/run

**Description**: Manually trigger workflow execution
**Response (201 Created)**:

```
{
  "success": true,
  "data": {
    "run_id": "run_456def",
    "workflow_id": "wf_123abc",
    "status": "running",
    "started_at": "2024-01-20T10:15:00Z"
  }
}
```

## 15.7 GET /workflows/{id}/runs

**Description**: Get workflow run history
**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "workflow_id": "wf_123abc",
    "runs": [
      {
        "id": "run_456def",
        "status": "completed",
        "started_at": "2024-01-19T09:00:00Z",
        "completed_at": "2024-01-19T09:15:00Z",
        "items_processed": 45
      }
```

```
      ]
    }
}
```

## 15.8 PATCH /workflows/{id}/toggle

**Description**: Enable or disable workflow

**Request Body**:

```
{
  "enabled": false
}
```

**Response (200 OK)**

# Chapter 16

# Trends Endpoints

## 16.1 GET /trends/

**Description**: Get technology trends with analysis

**Query Parameters**:

- `category`: Filter by category (ai, cloud, security, data)
- `period`: Time period (3m, 6m, 12m, default: 6m)

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "period": "6m",
    "category": "ai",
    "trends": [
      {
        "technology": "Large Language Models",
        "trend_score": 98,
        "trajectory": "up",
        "mentions_count": 2450,
        "growth_rate": 0.35
      }
    ]
  }
}
```

## 16.2 GET /trends/radar

**Description**: Get technology radar data for visualization

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "quadrants": {
```

```
      "techniques": [
        {"name": "Few-shot Learning", "ring": "trial"}
      ],
      "tools": [
        {"name": "LLaMA", "ring": "assess"}
      ],
      "platforms": [
        {"name": "Cloud GPU Services", "ring": "adopt"}
      ],
      "languages": [
        {"name": "Rust", "ring": "trial"}
      ]
    }
  }
}
```

## 16.3 GET /trends/timeline

**Description**: Get timeline for specific technology adoption

**Query Parameters**:

- `tech`: Technology name (required)

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "technology": "Kubernetes",
    "timeline": [
      {
        "year": 2015,
        "adoption_percentage": 2,
        "milestone": "Kubernetes 1.0 released"
      },
      {
        "year": 2024,
        "adoption_percentage": 68,
        "milestone": "Industry standard for orchestration"
      }
    ]
  }
}
```

# Chapter 17

# Collections Endpoints

## 17.1 POST /collections/

**Description**: Create custom collection for organizing content

**Headers**:

```
Authorization: Bearer <access_token>
Content-Type: application/json
```

**Request Body**:

```
{
  "name": "AI Safety Research",
  "description": "Papers and articles on AI safety",
  "is_public": false,
  "tags": ["ai", "safety", "alignment"]
}
```

**Response (201 Created)**:

```
{
  "success": true,
  "data": {
    "id": "col_abc123",
    "name": "AI Safety Research",
    "description": "...",
    "is_public": false,
    "item_count": 0,
    "created_at": "2024-01-20T10:00:00Z"
  }
}
```

## 17.2 GET /collections/

**Description**: List user's collections

**Query Parameters**:

- page: Page number

- `sort`: Sort by (name, created, updated)

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "page": 1,
    "total": 12,
    "results": [
      {
        "id": "col_abc123",
        "name": "AI Safety Research",
        "item_count": 24,
        "created_at": "2024-01-20T10:00:00Z"
      }
    ]
  }
}
```

## 17.3 GET /collections/{id}

**Description**: Get collection with all items

**Query Parameters**:

- `page`: Page number
- `page_size`: Items per page

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "col_abc123",
    "name": "AI Safety Research",
    "description": "...",
    "item_count": 24,
    "items": [
      {
        "id": "item_123",
        "content_id": "pap_xyz789",
        "content_type": "paper",
        "title": "...",
        "added_at": "2024-01-20T10:00:00Z"
      }
    ]
  }
}
```

## 17.4 PUT /collections/{id}

**Description**: Update collection metadata
**Response (200 OK)**

## 17.5 DELETE /collections/{id}

**Description**: Delete collection
**Response (204 No Content)**

## 17.6 POST /collections/{id}/items

**Description**: Add item to collection
**Request Body**:

```
{
  "content_type": "paper",
  "content_id": "pap_xyz789",
  "notes": "Important foundational paper on AI safety"
}
```

**Response (201 Created)**:

```
{
  "success": true,
  "data": {
    "id": "item_124",
    "content_id": "pap_xyz789",
    "added_at": "2024-01-20T10:05:00Z"
  }
}
```

## 17.7 DELETE /collections/{id}/items/{item_id}

**Description**: Remove item from collection
**Response (204 No Content)**

# Chapter 18

# Notifications Endpoints

## 18.1 GET /notifications/

**Description**: List user notifications

**Query Parameters**:

- `page`: Page number
- `unread_only`: Boolean, default false

**Headers**:

```
Authorization: Bearer <access_token>
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "page": 1,
    "total": 34,
    "unread_count": 5,
    "results": [
      {
        "id": "notif_abc123",
        "type": "new_paper",
        "title": "New paper matching your interests",
        "message": "A new paper on AI safety...",
        "read": false,
        "created_at": "2024-01-20T10:00:00Z",
        "data": {
          "paper_id": "pap_xyz789"
        }
      }
    ]
  }
}
```

## 18.2 PATCH /notifications/{id}/read

**Description**: Mark notification as read

**Headers**:

```
Authorization: Bearer <access_token>
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "notif_abc123",
    "read": true
  }
}
```

## 18.3 POST /notifications/read-all

**Description**: Mark all notifications as read

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "marked_read": 5
  }
}
```

# Chapter 19

# Admin Endpoints

## 19.1 GET /admin/stats

**Description**: Get platform-wide statistics (admin only)

**Headers**:

```
Authorization: Bearer <admin_token>
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "users": {
      "total": 15234,
      "active_this_month": 8923,
      "new_this_week": 234
    },
    "content": {
      "total_articles": 54000,
      "total_papers": 128000,
      "total_repositories": 45000
    },
    "engagement": {
      "avg_sessions_per_user": 8.5,
      "bookmarks_created_this_week": 12450
    }
  }
}
```

## 19.2 POST /admin/scrapers/trigger

**Description**: Manually trigger content scrapers

**Request Body**:

```
{
  "sources": ["arxiv", "github", "medium"],
  "force_refresh": true
```

```
}
```

**Response (202 Accepted)**:

```
{
  "success": true,
  "data": {
    "job_id": "scrape_12345",
    "status": "queued",
    "sources": ["arxiv", "github", "medium"]
  }
}
```

## 19.3 GET /admin/scrapers/status

**Description**: Get current scraper status and health

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "scrapers": [
      {
        "source": "arxiv",
        "status": "running",
        "last_update": "2024-01-20T10:15:00Z",
        "items_processed_today": 1250,
        "error_count": 0
      }
    ]
  }
}
```

## 19.4 GET /admin/users

**Description**: List all platform users (admin only)

**Query Parameters**:

- `page`: Page number
- `status`: Filter by status (active, inactive, suspended)
- `tier`: Filter by subscription tier

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "page": 1,
    "total": 15234,
```

```
    "results": [
      {
        "id": "usr_12345abcde",
        "email": "user@example.com",
        "name": "John Doe",
        "tier": "premium",
        "status": "active",
        "created_at": "2024-01-15T10:30:00Z"
      }
    ]
  }
}
```

## 19.5 PATCH /admin/users/{id}/role

**Description**: Change user subscription tier or role

**Request Body**:

```
{
  "tier": "enterprise",
  "role": "admin"
}
```

**Response (200 OK)**:

```
{
  "success": true,
  "data": {
    "id": "usr_12345abcde",
    "tier": "enterprise",
    "role": "admin",
    "updated_at": "2024-01-20T10:30:00Z"
  }
}
```

# Chapter 20

# WebSocket Endpoints

## 20.1 Real-time Updates

SYNAPSE provides WebSocket endpoints for real-time updates and live data streaming.

## 20.2 Connection

**WebSocket URL**:

```
wss://api.synapse.ai/ws/v1?token=<access_token>
```

**Connection Message**:

```
{
  "type": "subscribe",
  "channels": [
    "notifications",
    "trending_updates",
    "conversation_typing"
  ]
}
```

## 20.3 Message Types

### 20.3.1 Notification Updates

**Channel**: `notifications`

**Message**:

```
{
  "type": "notification",
  "data": {
    "id": "notif_abc123",
    "message": "New trending article",
    "created_at": "2024-01-20T10:30:00Z"
  }
}
```

### 20.3.2 Trending Updates

**Channel**: `trending_updates`

**Message**:

```
{
  "type": "trending_update",
  "data": {
    "technology": "Large Language Models",
    "current_trend_score": 98,
    "change_percentage": 5.2
  }
}
```

### 20.3.3 Chat Typing Indicator

**Channel**: `conversation_typing`

**Message**:

```
{
  "type": "typing",
  "data": {
    "conversation_id": "conv_abc123",
    "user_id": "usr_12345abcde",
    "is_typing": true
  }
}
```

# Chapter 21

# Summary and Best Practices

## 21.1 API Best Practices

1. **Always include Authorization header** for authenticated endpoints
2. **Implement exponential backoff** for rate limit retries
3. **Use pagination** for list endpoints to manage large datasets
4. **Monitor X-RateLimit headers** to avoid hitting rate limits
5. **Cache responses** where appropriate to reduce API calls
6. **Use semantic search** instead of keyword search for better results

## 21.2 Error Handling

Implement comprehensive error handling in your client:

- Check HTTP status codes and error response structure
- Implement retry logic with exponential backoff for 5xx errors
- Handle 401 errors by refreshing the access token
- Log API errors for debugging and monitoring

## 21.3 Pagination Best Practices

- Start with default page size and adjust based on performance
- Use cursor-based pagination for large datasets if available
- Cache pagination state on client side
- Never assume page count remains constant